

High Steaks: Modeling the Long-Term Temperature Impact of Short- & Long-Lived Greenhouse Gas Emissions from the U.S. Beef Industry

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Introduction

- ❖ Beef production in the United States is a fifth of the global industry (Rotz 2019), and its greenhouse gas (GHG) emissions is concerning in a world impacted by climate change
- ❖ Past research have used Life Cycle Assessment (LCA) models to estimate and predict cattle emissions; environmental impact is often quantified in terms of CO₂ equivalents (CO₂e)
- ❖ Some climate researchers have proposed that CO₂e is an inadequate metric for global warming by focusing on pulse emissions and “destroys information needed to make a proper assessment of climate damage” (Pierrehumbert and Eshel 2015)
- ❖ We attempt to expand on the work of Pierrehumbert and Eshel by using a more complex set of models to forecast temperature perturbation due to the U.S. beef industry; we also examined the difference in perturbations between the current beef system and one that relies mostly on grass-finishing.

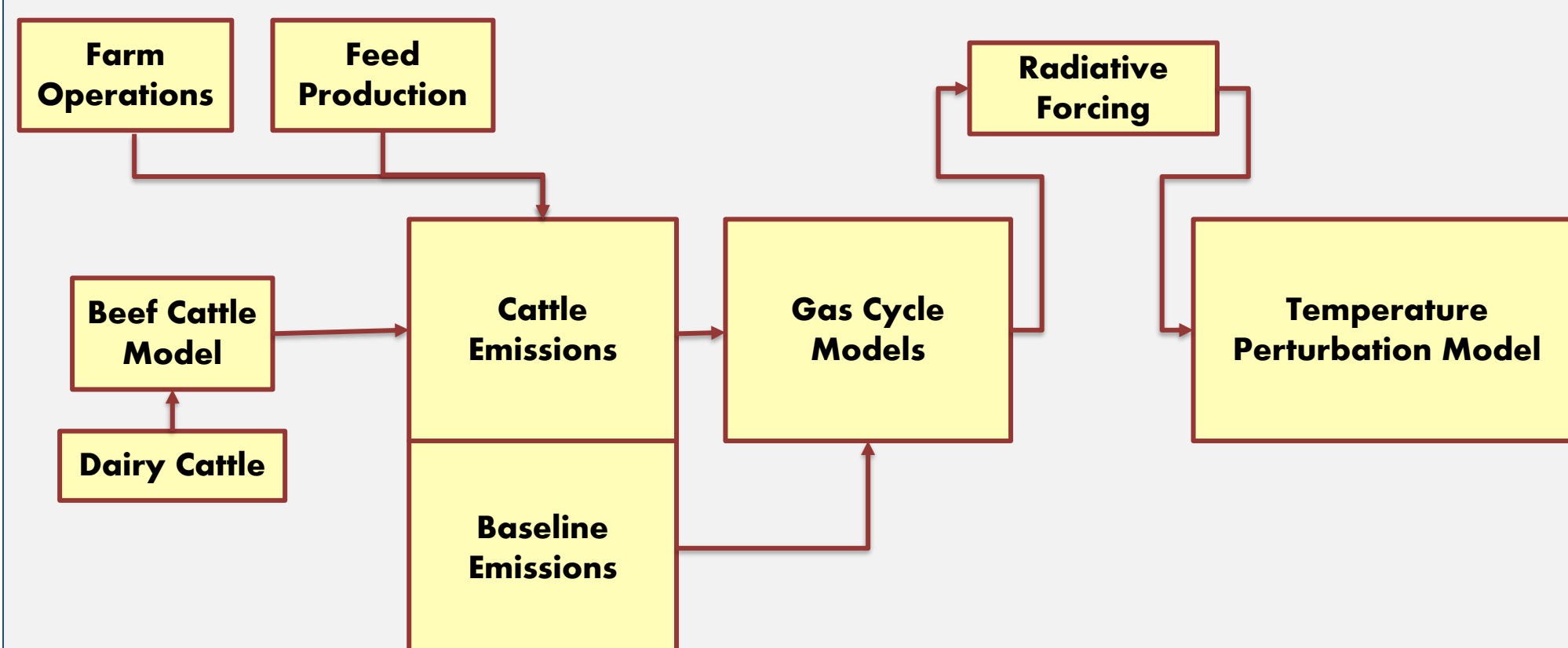
Research Questions

We model the temperature perturbation from beef industry emissions to:

- Observe the long-term warming due to the industry’s emissions
- Model changes in the industry’s practices and determine the associated change in global temperature
- Evaluate the accuracy of CO₂e projections on a time-scale of hundreds of years

Methodology

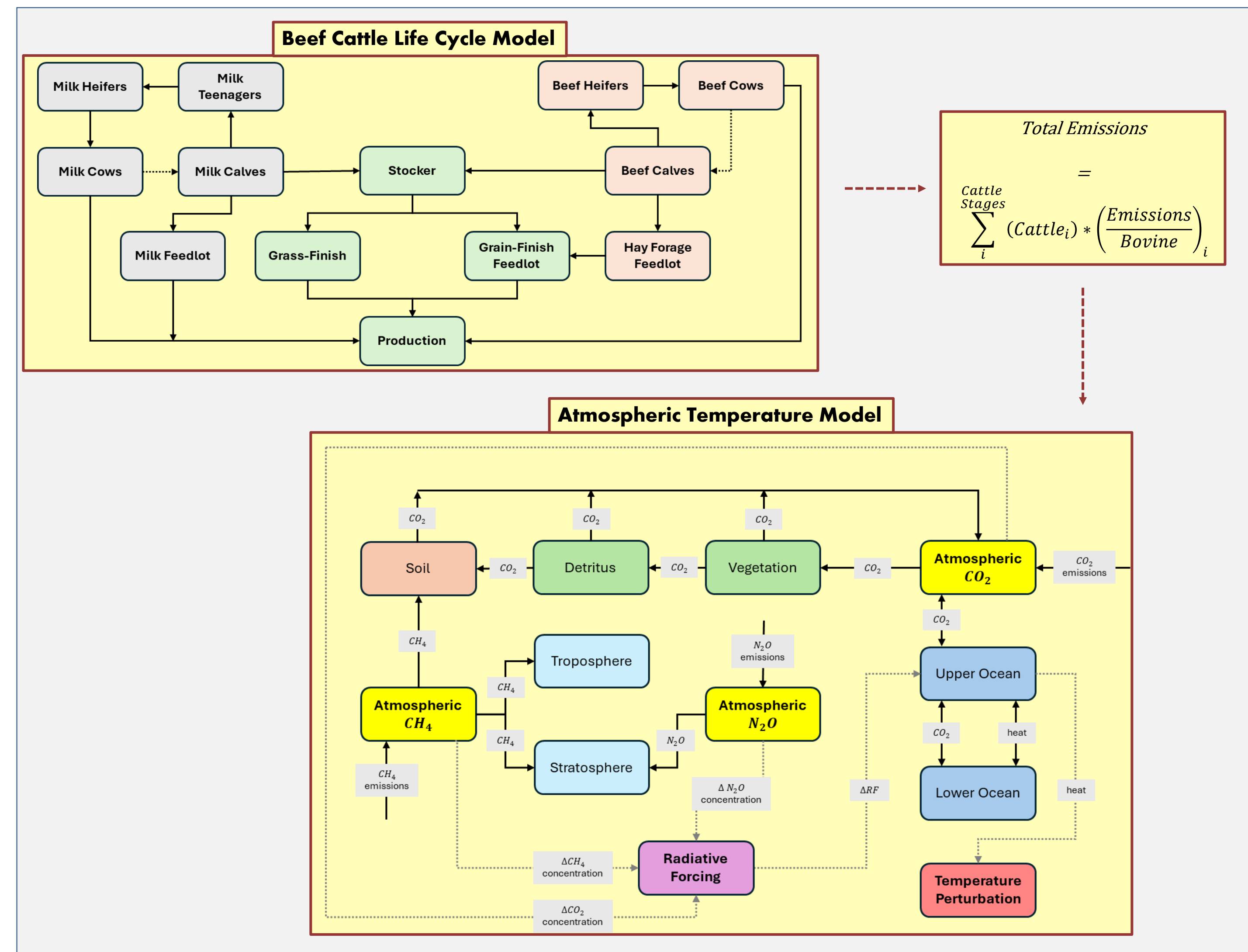
We use 2 systems of ordinary differential equations to create an LCA model of U.S. beef cattle as well as a reduced-complexity climate model. The total GHG emissions per USDA region is inputted into the climate model to assess the impact on global temperature perturbation.



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Mathematical Model



Model Results



Conclusions

- ❖ Model projects a temperature perturbation of 0.0125 K due to the U.S. beef industry in 2500
- ❖ Grass-finished beef produces more GHG emissions compared to grain-finished beef
- ❖ On a time-scale of hundreds of years, CO₂e is an inadequate metric for global warming

Limitations

- ❖ Model would benefit from more rigorous parameter-estimation
- ❖ Data regarding regional cattle industry differences are very limited
- ❖ Dairy industry emissions are not yet considered

Future Scope

- ❖ More research into N₂O mitigation methods for the cattle industry should be done
- ❖ Dairy cattle emissions should be included in future research to obtain a fuller picture of the situation

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